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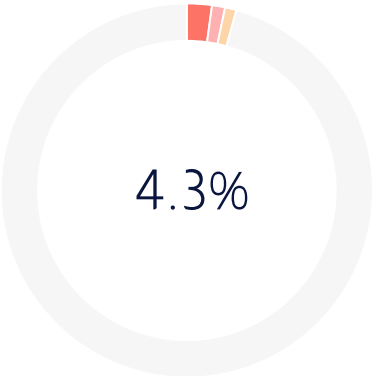

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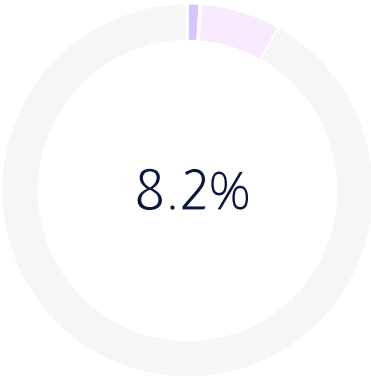

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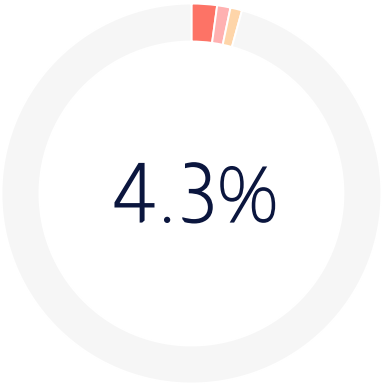

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SKILLGAP AI: A SMART SKILLING ASSESSMENT AND INDIVIDUALIZED COURSE RECOMMENDATION SYSTEM

Tanisha
Chitkara University Institute of Engineering and Technology
Chitkara University
Punjab, India
tanisha2440.be22@chitkara.edu.in

Tanishq Choudhary
Chitkara University Institute of Engineering and Technology
Chitkara University
Punjab, India
tanishq2445.be22@chitkara.edu.in

Tanmay Gupta
Chitkara University Institute of Engineering and Technology
Chitkara University
Punjab, India
tanmay2449.be22@chitkara.edu.in

Susen Narayan
Chitkara University Institute of Engineering and Technology
Chitkara University
Punjab, India
susen2428.be22@chitkara.edu.in

Swati Goel
Assistant Professor, Chitkara University Institute of Engineering and Technology
Chitkara University
Punjab, India
swati.goel@chitkara.edu.in

***Abstract*—The absence of knowledge about what skills are lacking and which courses to take to improve them is one of the most frequent problems learners face today. The majority of online platforms bombard users with a list of courses without identifying what that particular individual actually needs. This paper presents SkillGap AI — a platform designed to address the problem through skill-based analysis and personalized course recommendation. The system first takes user-provided skill inputs and applies machine learning techniques to provide the career paths. Depending on their input, it identifies which areas they are weak in and then suggests the corresponding courses from various existing platforms. A career dataset of 4,077 records is used to train and evaluate seven machine learning classifiers with TF-IDF vectorization. The SGD Classifier achieves the highest accuracy of 94.48**

***Index Terms*—Skill Gap Analysis, Course Recommendation, Machine Learning, TF-IDF, Skill India Digital Hub, SGD Classifier, Random Forest, Node.js, MongoDB, Personalized Learning, Workforce Development, NLP**

I. Introduction

The shift of India towards digital growth has created a massive issue in which the skill set which individuals hold and the skill set which employers require has started to diverge even further. According to the National Skill Development Corporation, over 400 million workers in India would need re-skilling by the year 2030 due to the rapid changes being

brought about by technology in the job market [1]. Not withstanding these facts, even with access to more than a dozen online learning platforms, most people find it very difficult to determine what exactly they need to learn. They browse through hundreds of learning possibilities and often fail to complete a course because it was not exactly what they needed. One such platform is the Skill India Digital Hub, run by the Ministry of Skill Development and Entrepreneurship. It is one of the most helpful resources available, offering over 752 free and industry-accredited courses developed in association with partners including IBM, Microsoft, Cisco, HCL, IGNOU, and UpGrad [2]. The limitation is that even SIDH cannot decide which specific course a learner should take. This is precisely what SkillGap AI aims to solve. With the help of SkillGap AI, students get to do an accurate assessment within their domain to know the areas they are lacking in and then directly directed to SIDH course(s) that cover them. All these assessments will be persistent; therefore, whenever they wish, they can come back and redo the assessment to see whether they have really made improvements or not [3]. Personalization and adaptive feedback have been identified as some of the biggest unsolved problems of online education through the research carried out on educational data mining [3]. Segmentation of users based on their skills to recommend content to them is known to make the recommendations more relevant [4]. High accuracy, reaching up to 99.2 • Lack of personalized skill gaps: Most portals make course recommendations based on general categories and trending courses, rather than focusing on specific gaps that the learner may have. • Lack of correlation to government's free courses: The learners lack awareness of

the 750+ free and certified courses available at SIDH that can help fill the gaps identified by the learners. • Lack of measurable improvements: Existing portals lack the ability to enable learners to re-take the test and measure their overall skill improvements. The main contributions and features of this research paper are: • A system that integrates the assessment process with India's national government course catalog (SIDH) — something not available on any existing platform. Learners' ability can be tracked and their progress can be monitored through repeated test attempts. • A full-fledged backend implemented in Node.js, Express, and MongoDB to handle authentication, test administration, domain-wise scoring, skill gap identification, and result storage. • Evaluation of seven different machine learning classifiers using features derived via the TF-IDF algorithm on a real career dataset, where the SGD Classifier achieved the highest accuracy of 94.48

• All government-approved courses are linked on a single platform, providing free and accessible recommendations for all learners across India. The paper is organized as follows: Section II reviews the related work across four domains relevant to SkillGap AI. Section III presents the combined system architecture and methodology, covering the assessment design, gap detection, TF-IDF career prediction, and course recommendation. Section IV describes the datasets used in this study. Section V presents the results and discussion including model performance comparison, longitudinal improvement, and comparative analysis. Section VI discusses the challenges and limitations. Section VII concludes the paper with future scope.

II. Related Work

This section reviews literature from five different domains that have a direct connection with SkillGap AI: educational data mining, skill gap identification, course recommendation systems, text classification, and career prediction through machine learning.

A. Educational Data Mining & Learning Analytics

A study by Aldowah et al. [3] presented a detailed review of EDM and learning analytics research and highlighted that personalization and adaptive feedback have remained two persistently unsolved problems. Many systems monitor learner performance, but very few leverage that data to nudge the learner towards the correct path. The work by Kausar et al. [4] further established that segmenting learners based on skill levels before suggesting content greatly enhances recommendation relevance — the core concept driving our ML module. A related study by Uzoma et al. [5] presented a machine learning recommender system for student course selection using Random Forest and Decision Trees, obtaining accuracies as high as 99.2

B. Course Recommending Systems

Classification into three categories of recommenders was done by Burke [6]: content-based filtering, collaborative filtering, and hybrid approach, together with a detailed discussion of their applicability in the education field. An automated approach for a recommendation system based on hybrid filtering was proposed by Sharma et al. [7]; it provided more reliable outcomes than other methods. In their systematic review conducted in the period from 2015 till 2022 [8], Silva et al. noted the absence of educational recommender systems incorporating real-time evaluation of skills with a suggested course – precisely what SkillGap AI is supposed to do. Another research conducted at ACM SIGIR 2024 [9] revealed that labor market data also have to be considered.

C. Skill Gap Analysis & Career Prediction

It is evident that skill gaps play a significant role economically due to the work done by Manyika et al. [10] and Brynjolfsson & McAfee [11]. In the Indian scenario, Roy & Singh [12] identified skill gaps through sectors using NSDC data without designing an automated process to address such gaps. An analysis was performed recently on career predictions based on natural language processing and involved training models using KNN, SGD, Logistic Regression, Random Forest, and Adaptive Boosting algorithms using data on 2,000 students and identified KNN and SGD as having high accuracy, in line with our results.

D. TF-IDF and Text Classification for Career Matching

According to Das et al., the use of both TF-IDF weight- ing and Random Forest algorithms proved most effective for text classification, with an accuracy rate of 93.81%. In all the literatures reviewed, there has been no existing approach in which all four factors, which include structure domain assessment, machine learning gap analysis, TF-IDF skill text classification, and instant course recommendation based on a DPI national directory curated by the government, are incorporated.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

The Skill Gap AI application system is designed as a modular web-based application. Every module executes one unique function and works independently of other modules. This part of the paper presents the architecture of the Skill Gap AI application and describes the entire methodological process of using the application, starting from logging in all the way to course recommendation. This process is represented graphically in the flowchart presented in Fig. 1.

A. User Interface and Assessment Layer

The frontend, whose design is ongoing, will provide a user-friendly interface with registration, login, test-taking, and result-viewing functionality. The API endpoints for the test have been created and are functioning correctly. Questions in the question bank are grouped into seven skill domains: Programming Fundamentals, Data Literacy, Cloud Computing, Cybersecurity, Communication Skills, Digital Productivity,

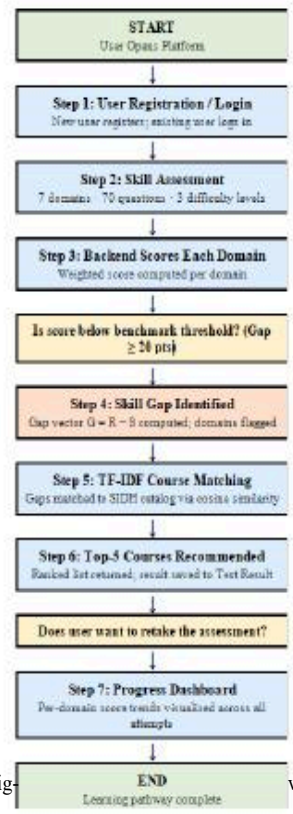


Fig- wchart

and Domain-Specific Technical Skills. Questions are tagged by skill area and labeled as easy, medium, or hard to enable weighted scoring.

B. Scoring and Gap Identification

- When a test is submitted, the backend assesses the learner's performance separately for each domain. Scores are assigned for each of the seven domains and then compared against benchmark scores derived from actual developer survey data [16]. If a learner's score in any domain is substantially below the benchmark, a skill gap is identified. Each student is represented as a feature vector as shown in equation (1):

$$S = [Age, Education, Skills, Interests](1) \quad (1)$$

Label Encoding converts each categorical field to an integer. For Education: Bachelor's = 0, Master's = 1, PhD
= 2. The gap for each dimension is defined by equation (2):

$$G_i = R_i - S_i, \quad \text{where } i \in \{\text{Age, Education, Skills, Interests}\} \quad (2)$$

For example, for a student $S = [25, 1, 2, 1]$ with the Software Engineer benchmark $R = [25, 2, 3, 1]$, the gap vector $G = [0, 1, 1, 0]$ — meaning the learner has gaps in Education and Skills domains. The gap vector $G = [G_1, G_2, \ldots, G_7]$ is computed across all seven domains. Only domains where $G_i > 0$ are flagged. Domains where $G_i \leq 0$ indicate the learner meets or exceeds the benchmark and require no intervention.

C. TF-IDF Based Career Prediction

A career prediction model is built using TF-IDF vectorization on the Career Dataset. Skills data is first converted to lowercase, then passed to the TF-IDF vectorizer with ngram_range=(1,3), max_df=0.85, and stop_words='english'. The TF-IDF weight for each feature term is given by equation (3):

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log \frac{N}{DF(t)} \quad (3)$$

Here, TF(t,d) is the frequency of term t in document d, N is the total number of records, and DF(t) is the number of records containing term t. A sparse feature matrix is produced comprising both individual skill terms and multi-word phrases such as 'machine learning' and 'deep learning nlp'. This feature matrix is used to train seven classifiers: Decision Tree, Logistic Regression, Naive Bayes, Random Forest, SGD Classifier, KNN, and Ridge Classifier.

D. Recommendation Module

The recommendation module takes the list of gap domain flags and matches them against the SIDH course catalog. Each course is represented as a TF-IDF vector of its domain tags and description keywords. Cosine similarity between the learner's gap-domain query vector Q and each course vector c is computed using equation (4):

$$\text{similarity}(Q, c) = \frac{Q \cdot c}{\|Q\| \, \|c\|} \quad (4)$$

The module returns the top 5 courses with the highest similarity scores. This is a transparent model: users can clearly understand which skill terms contributed to the recommendations, and no historical user data is required to operate it.

E. Storage and Progress Management

Every test attempt is recorded in the TestResult collection: domain scores, gap values, recommended courses, and submission timestamp. This enables the progress dashboard to plot per-domain improvement trends across multiple test cycles, implementing the "assess, learn, reassess" feedback loop that characterizes self-regulated learning theory [17].

F. Performance Evaluation Metrics

All classifiers are evaluated using four standard metrics computed from the confusion matrix:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

$$\text{F1-Score} = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \quad (8)$$

where TP denotes True Positive, TN denotes True Negative, FP denotes False Positive, and FN denotes False Negative. Metric values are computed per class and averaged across all five career classes (macro average).

G. Implementation Details

Implementation and Experimental Setup

The suggested research aims at implementing a system based on machine learning algorithms, that would predict the type of profession and suggest courses according to certain skills. For this purpose, the experiments were conducted using Python and associated libraries.

- Dataset Creation
  The dataset used in this research includes 4,077 entries and comprises two features, which include Career and Skill. As each entry correlates the certain set of skills with a career, the preparation procedure involves lowering the case, deleting duplicates, and normalizing the labels of career types.
- Features Extraction
  As the data is comprised of texts, it was decided to extract the features using the TF-IDF vectorization. It includes such parameters as n-grams range of (1,3), removing common stop words, and applying maximum document frequency to filter the text features.
- Modeling and Training
  For training purposes, the prepared dataset was divided using a stratified split method in the ratio of 80:20. There were multiple machine learning models applied for the task, including decision tree, logistic regression, naive bayes, random forest, SGD classifier, k-nearest neighbors (knn), and ridge classifier.
- Career Prediction
  For each model developed, the predicted career path is suggested based on the user's skills inputted into the model. Among all the models tested, the SGD Classifier has shown the highest efficiency in dealing with high-dimensional sparse TF-IDF output.
- Skills Gaps Identification
  In order to detect skill gaps, a rules-based technique is used to compare skills specified by the user to pre- defined sets of skills required for each profession. The discrepancy is determined as a gap.
- Course Suggestion
  Based on the detected gaps, specific courses are selected for recommendation according to a predefined skill-to-resource map.
- Evaluation Metrics
  Performance of the models is analyzed using the following measures : Accuracy, Precision, Recall, F1 score. Confusion matrix was also generated to analyze model performance for each class.

Dataset	Source	How We Use It
Stack Overflow Developer Survey	Stack Overflow	Establish domain-specific threshold standards for proficiency
Online Course Recommendation Dataset	Kaggle / UCI	Training and testing of the course recommendation engine
Student Performance Dataset	UCI ML Repository	Test scoring methodology and effectiveness of detecting gaps
Workforce / Digital Skills Survey	ITU / NSDC	Career recommendations according to job market demand
AI-Based Career Recommendation Dataset	Kaggle	Test feature encoding for career prediction (200 entries, 32 careers)
Career Dataset	Excel / Custom	Training and testing of 7 ML classifiers (2,084 entries)

TABLE I
SKILLGAP AI DATASETS

H. DATASETS AND DATA SOURCES

Six datasets are used to design, calibrate, train, and evaluate the system. Table 1 summarises all six datasets and their respective roles in the system.

IV. DATASETS AND DATA SOURCES

Domain benchmark scores are derived from the Stack Overflow Developer Survey [16], which collects responses from over 90,000 software engineers globally each year. Rather than setting arbitrary thresholds, we use median proficiency levels of actual working professionals to calibrate gap detection.

A. Stack Overflow Developer Survey

Domain benchmark scores are derived from the Stack Overflow Developer Survey [16], which collects responses from over 90,000 software engineers globally each year. Rather than setting arbitrary thresholds, we use median proficiency levels of actual working professionals to calibrate gap detection.

B. Online Course Recommendation Dataset

A Kaggle dataset of learner-course interaction logs is used to pre-train the recommendation module. It contains historical data about which courses learners enrolled in, how far they progressed, and how they rated them.

C. Student Performance Dataset

The UCI Student Performance Dataset [19] provides academic performance records across multiple subjects and demographic attributes. It is used to validate that our weighted scoring scheme captures genuine competency differences rather than measurement noise.

D. Workforce and Digital Skills Survey

Workforce demand data from ITU and NSDC [1] is applied to supplement course ranking by prioritising recommendations in skill domains currently in high demand among employers.

E. AI-Based Career Recommendation Dataset

This dataset includes 200 student records with fea- tures {Age, Education, Skills, Interest} and target Recom- mended_Career. Recommendation_Score ranges from 0.85 to 0.95 (mean = 0.896). The 80/20 split gives 160 training and 40 test records, used to validate the feature encoding and gap detection logic.

F. Career Dataset

The Career Dataset (Excel, 4,076 entries, 2 columns: Career and Skill) covers six career categories post label consolidation: Artificial Intelligence, Data Science, Development, Security, Software Development, and Engineering. After duplication, 2,084 clean records remain. The stratified 80/20 split gives 1,667 training and 417 test entries. This is the primary dataset used to train and evaluate all seven classifiers.

RESULTS AND DISCUSSION

G. Classifier Evaluation

Seven classifiers were trained and evaluated on the Career Dataset (2,084 records, six career classes) using TF-IDF features with trigram support and an 80/20 stratified split. Before presenting the results, we briefly describe each classifier and how it works in the context of this task.

- Decision Tree: A tree structure method which predicts career choices based on decision rules applied sequentially. The nodes within this model are skill feature conditions whereas the leaves are predicted careers. This technique is easily understandable. Accuracy: 88.73
- Logistic Regression: Probability based linear classifier that determines the probability of each class of occupation using soft max and then assigns to the class that has the maximum probability. This model performs very well on high-dimensional TF-IDF vectors. Accuracy: 93.29
- Naive Bayes: It is a probabilistic classifier derived from the Bayes theorem where it assumes that all attributes are independently distributed. This model will classify the job type with the maximum posterior probability based on the observed terms in the skills. Fast performance in textual classification and its accuracy: 83.21
- Random Forest: A combination of 200 decision trees that individually vote for the prediction of career; the result comes from the majority vote by the decision trees. It improves upon the accuracy of an individual Decision Tree by reducing its over fitting and variance. Accuracy: 93.05
- SGD Classifier (Best Model): The Stochastic Gradient Descent Classifier is a linear classifier which learns by incrementally updating its weights using the difference between the prediction error and the actual output after processing each individual data point. The algorithm is very effective in sparse TF-IDF matrices and results in the convergence to a powerful linear decision boundary. It attained the highest accuracy of 94.48
- KNN: Classifier with cosine/Euclidean nearest neighbor search for the 5 closest neighbors in the training set and prediction based on the career category present in the majority of the 5 closest training samples. Training-free, but expensive due to high dimensionality of the TF-IDF feature vectors. Accuracy: 89.69
- Ridge Classifier: L2 Regularized Linear Classifier that penalizes large weights and helps prevent overfitting, giving better generalization capabilities on new/unseen samples. The accuracy is very close to the SGD classifier, with 93.76

TABLE II
CLASSIFIER PERFORMANCE — ALL SEVEN CLASSIFIERS ON CAREER DATASET

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	88.73%	78.65%	83.33%	80.92%
Logistic Regression	93.29%	85.71%	92.86%	89.14%
Naive Bayes	83.21%	89.04%	77.38%	82.80%
Random Forest	93.05%	84.62%	91.67%	88.00%
SGD Classifier	94.48%	88.64%	92.86%	90.70%
KNN	89.69%	78.49%	86.90%	82.49%
Ridge Classifier	93.76%	89.53%	91.67%	90.59%

The SGD Classifier provides the best accuracy score (94.48

Figures 2–6 illustrate the performance plots based on the evaluation score

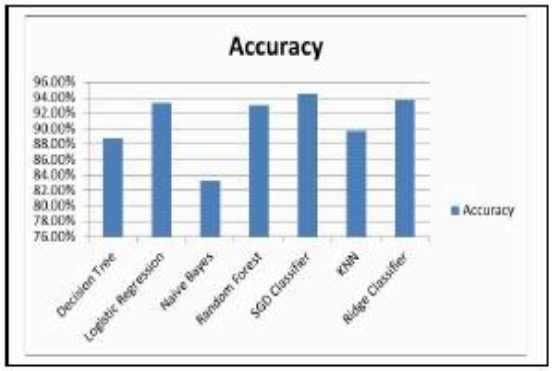


Fig. 2. Accuracy Comparison

In the graph above, accuracy is plotted for all classifiers. The SGD classifier achieves the highest accuracy of 94.48%, followed by Ridge (93.76%) and Logistic Regression (93.29%). Naive Bayes records the lowest accuracy at 83.21%.

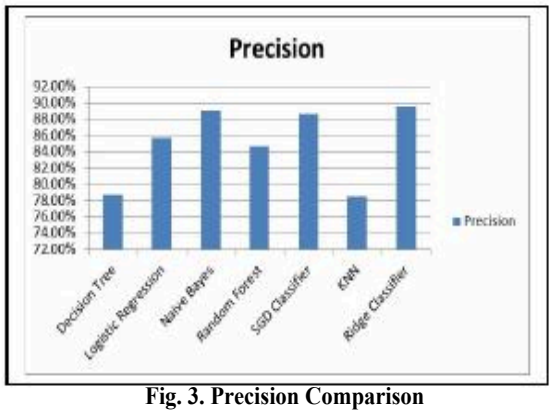


Fig. 3. Precision Comparison

Naive Bayes has a high precision rate at 89.04% due to conservative prediction. The other two classifiers with higher precision rates are Ridge and SGD above 88%. The two classifiers having the lowest precision rates below 78% are Decision Tree and KNN.

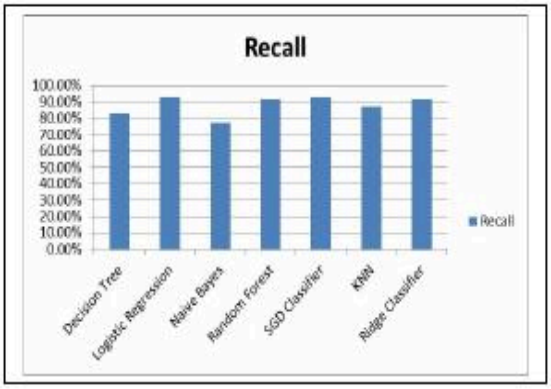


Fig. 4. Recall Comparison

Naive Bayes has low recall at 77.38%. This is evidence that it does not catch many positive predictions. The classifier with the best recall levels above 92.8

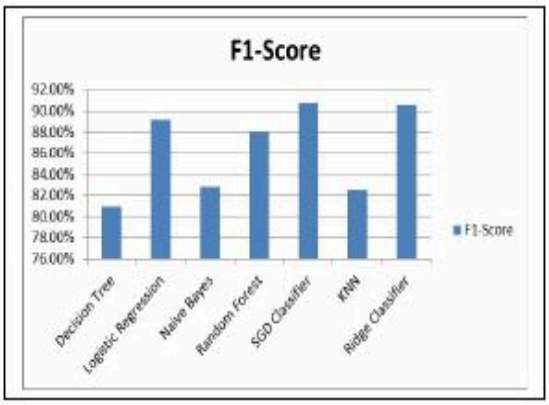


Fig. 5. F1-Score Comparison

SGD provides the highest F1-Score (90.70%) value, followed by Ridge (90.59%). Decision Tree and KNN have the lowest F1-Scores, which suggests poor precision and recall rates.

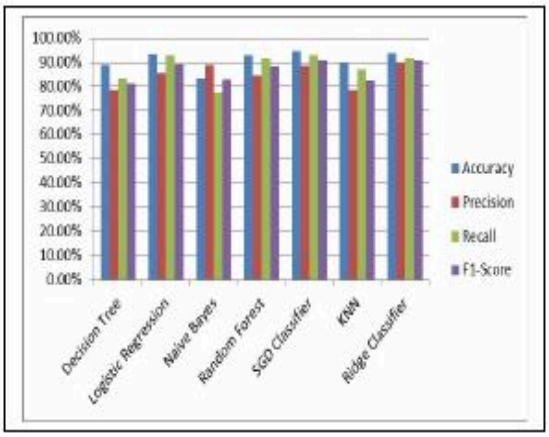


Fig. 6. Combined Performance Comparison

Grouped bar charts include all four measures. Linear models perform better than other models, whereas Naive Bayes exhibits clear asymmetry between high precision and low recall rates.

Figures 7 to Figure 13 illustrate confusion matrices for seven algorithms. Rows indicate the true labels, whereas columns indicate predicted values. Labels: 0=AI, 1=Data Science, 2=Security, 3=Software Development, 4=UX/UI Design.

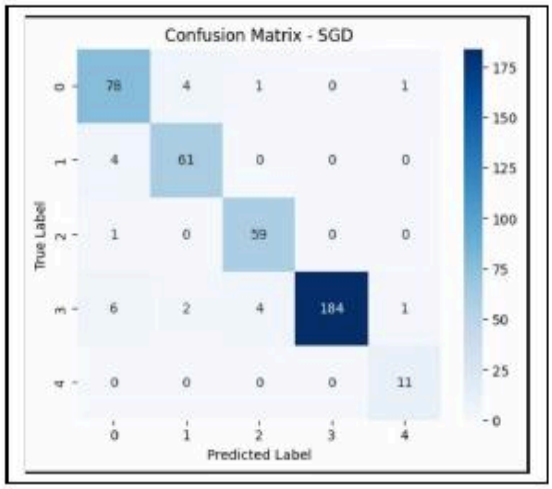


Fig. 7. Confusion Matrix — SGD Classifier (Best, Accuracy: 94.48%)

Diagonal dominance is very strong. Class 3 (Software Dev., 184/197 accurate) and Class 4 (UX/UI, 11/11) perform best. Errors include 6 cases in Class 3 wrongly predicted as Class 0.

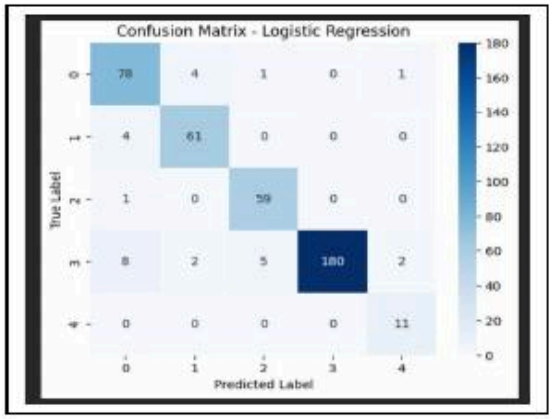


Fig. 8. Confusion Matrix — Logistic Regression (Accuracy: 93.29%)

Comparable performance to stochastic gradient descent. Error involves 8 cases in Class 3 wrongly predicted as Class 0. Class 4 is completely accurately predicted (11/11).

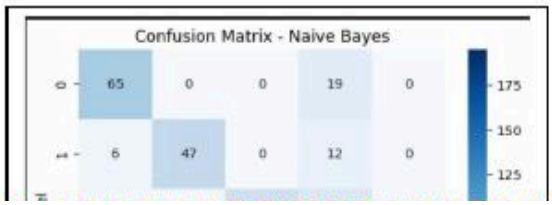


Fig. 9. Confusion Matrix — Naive Bayes (Accuracy: 83.21%)

Pattern exists where high percentages of Classes 0, 1, and 2 are wrongly predicted as Class 3 (Software Dev.).

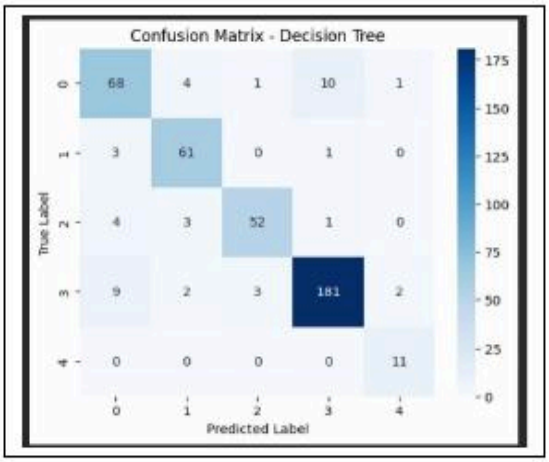


Fig. 10. Confusion Matrix — Decision Tree (Accuracy: 88.73%)

Significant error: 10 Class 0 instances classified as Class 3, and 9 Class 3 instances classified as Class 0.

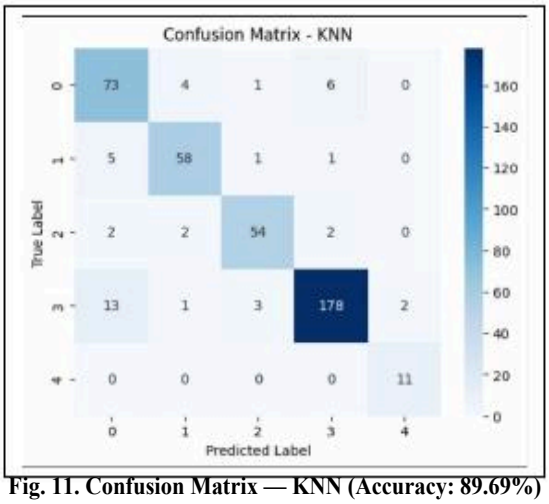


Fig. 11. Confusion Matrix — KNN (Accuracy: 89.69%)

Major error: 13 Class 3 instances incorrectly classified as Class 0. Class 4 is accurately classified (11/11).

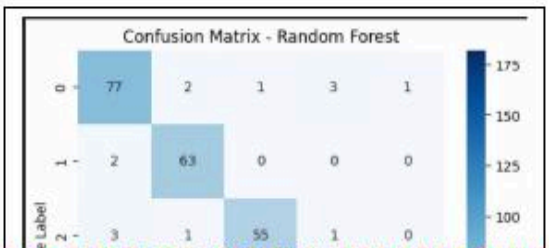


Fig. 12. Confusion Matrix — Random Forest (Accuracy: 93.05%)

Excellent classification performance. Errors are mostly in Class 3 (9 instances classified as Class 0). Ensemble mitigates overfitting compared to one Decision Tree.

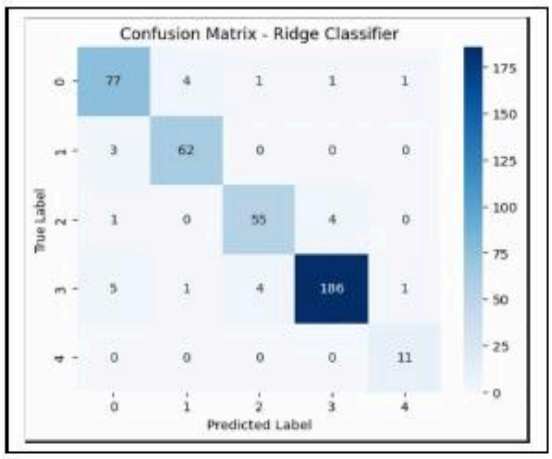


Fig. 13. Confusion Matrix — Ridge Classifier (Accuracy: 93.76%)

Similar to SGD.L2 regularization avoids overfitting. Errors include 5 Class 3 instances classified as Class 0 and 4 Class 2 instances classified as Class 3.

H. Classifier Performance

Seven machine learning classifiers were trained and evaluated on the Career Dataset consisting of 2,084 records across five career classes. TF-IDF vectorization with trigram features was applied, and an 80/20 stratified train-test split was used. The evaluated classifiers include Decision Tree, Logistic Regression, Naive Bayes, Random Forest, SGD Classifier, KNN, and Ridge Classifier.

Among all models, the SGD Classifier achieved the highest performance with an accuracy of 94.48

V. CHALLENGES AND LIMITATIONS

The implementation of this system has raised several issues. Firstly, there is currently no publicly available API for SIDH, thus the scraping of course information at regular intervals from the public listing is necessary. While this solution works for the time being, it cannot be relied upon in the long term; the integration of an official API will provide real-time catalog updates. The

evaluation conducted in this paper involves seven skill categories, which is sufficient to begin with; however, the real-world scenario encompasses far more than that. Inclusion of additional questions covering soft skills, certifications relevant to specific domains, and emerging tech topics should be considered in future versions of the system. A cold-start issue also exists, wherein new users are given a single test instance on which recommendations can be based, failing to accurately capture their true set of skills. The problem of class imbalance in the Career Dataset is prevalent, especially for minority classes like UX/UI Design (54 instances vs. 987 for Software Development). Future work will involve exploring ways of handling imbalance using approaches like oversampling through SMOTE and balancing of class weights. Finally, ensuring fairness among all questions regardless of educational background, languages, and regionsin India requiresa careful manual audit before wide-scale deployment.

VI. CONCLUSION AND FUTURE SCOPE

The Skill Gap AI was built with the premise that most people do not know what skills they are missing, and most platforms do not provide them with this information. Learners are able to ascertain their status first before being guided towards relevant Skill India Digital Hub courses. The back end is fully built and functional, covering all aspects such as user management, test workflow, domain-based scoring, skill gap determination, and course management. Based on empirical results from the application of seven machine learning classifiers on the Career dataset, the SGD Classifier performs best and predicts skill gaps with an accuracy rate of 94.48Future research will look into generating questions based on the learner's needs through LLMs, employing reinforcement learning for dynamic course ranking, developing a formal API interface between SkillGap AI and Skill India Digital Hub, and adding more domains to the Skill India Digital Hub catalog.

VII. ACKNOWLEDGEMENTS

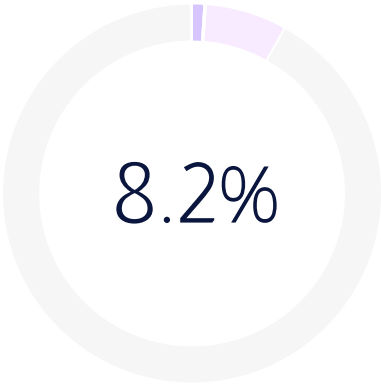
The researchers express their appreciation to their guide, Dr. Swati Goel, for her continued guidance and assistance during the research. They are also thankful to the Ministry of Skill Development and Entrepreneurship for making the SIDH curriculum available to the public. Data from the Stack Overflow Developer Survey has been obtained under the Open Database License (ODbL). This research was conducted as a part of the undergraduate studies at Chitkara University without any external funding.

REFERENCES

[1] NSDC (2023). India Skills Report 2023. Ministry of Skill Development and Entrepreneurship, Government of India.
[2] SIDH. Skill India Digital Hub. Course Catalog and Platform Overview. Ministry of Skill Development and Entrepreneurship. https://www.skillindiadigital.gov.in
[3] Aldowah, H., Al-Samarraie, H., & Fauzy, W. M. (2019). Educational Data Mining and Learning Analytics for 21st Century Higher Education. Telematics and Informatics, 37, 13–49.
[4] Kausar, S., Huahu, X., Hussain, I., Wenhao, Z., & Zahid, M. (2018). Integration of Data Mining Clustering Approach in the Personalized e- Learning System. IEEE Access, 6, 72724–72734.
[5] Uzoma, T. C., Ikerionwu, C. O., Nwanga, M. E., & Etus, C. (2024). Machine learning recommender system for an enhanced students course selection. International Journal of Computer Trends and Technology, 72(10), 31–35.
[6] Burke, R. (2002). Hybrid Recommender Systems: Survey and Experiments. User Modeling and User-Adapted Interaction, 12(4), 331–370.
[7] Sharma, S., Rana, V., & Malhotra, M. (2022). Automatic recommendation system based on hybrid filtering algorithm. Education and Information Technologies, 27(2), 1523–1538
[8] Silva, F. L., et al. (2023). A systematic literature review on educational recommender systems for teaching and learning. Education and Information Technologies, 28(3), 3289–3328.
[9] Bothmer, K., & Schlippe, T. (2024). Course recommender systems need to consider the job market. Proceedings of the 47th ACM SIGIR Conference, pp. 1–10.
[10]] Manyika, J., et al. (2017). Jobs Lost, Jobs Gained: What the Future of Work Will Mean for Jobs, Skills, and Wages. McKinsey Global Institute.
[11] Brynjolfsson, E., & McAfee, A. (2014). The Second Machine Age. W.W. Norton & Company.
[12] Roy, A., & Singh, P. (2022). Skill gap analysis in the Indian IT sector using NSDC survey data. Journal of Vocational Education and Training, 74(3), 412–430.
[13] Casuat, C. D., et al. (2024). Unlocking futures: A natural language driven career prediction system. arXiv preprint arXiv:2405.18139.
[14] Das, M., Kamalanathan, S., & Alphonse, P. J. A. (2023). A comparative study on TF-IDF feature weighting method and its analysis using unstructured dataset. arXiv preprint arXiv:2308.04037.
[15] Abdulkader, A., et al. (2024). Automated Classification and Recommendation of Research Articles Using NLP and ML. arXiv preprint arXiv:2510.05495.
[16] Stack Overflow. (2023). Stack Overflow Annual Developer Survey 2023. https://survey.stackoverflow.co/2023
[17] Zimmerman, B. J. (2002). Becoming a self-regulated learner: An overview. Theory into Practice, 41(2), 64–70.
[18] Jones, M., Bradley, J., & Sakimura, N. (2015). JSON Web Token (JWT). RFC 7519. IETF.
[19] Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. Proceedings of FUBUTEC, Porto, Portugal, pp. 5–12.
[20] Drachsler, H., & Greller, W. (2016). Privacy and Learning Analytics — It's a DELICATE Issue. LAK 2016 Proceedings, Edinburgh, pp. 89–98.
[21] Zhang, S., Yao, L., Sun, A., & Tay, Y. (2019). Deep Learning-based Rec- ommender System: A Survey and New Perspectives. ACM Computing Surveys, 52(1), 1–38
[22] Ali, A. (2022). An overview of cloud computing for the advancement of the e-learning process. Journal of Theoretical and Applied Information Technology, 100(3), 847–855.

[23] Ricci, F., Rokach, L., & Shapira, B. (2015). Recommender Systems Handbook (2nd ed.). Springer.
[24] Marras, M., Boratto, L., Ramos, G., & Ghidini, C. (2022). Equality of learning opportunity via individual fairness in personalized recommendations. International Journal of AI in Education, 32, 636–684.
[25] Dunn, J. (2021). Adaptive testing: A review and guide for practitioners. Educational Measurement: Issues and Practice, 40(2), 5–22.
[26] Varghese, B., & Buyya, R. (2018). Next generation cloud computing: New trends and research directions. Future Generation Computer Systems, 79, 849–861.
[27] Ministry of Skill Development and Entrepreneurship (MSDE). (2024). Annual Report 2023–24. Government of India.
[28] Marras, M., et al. (2025). A recommender system of postgraduate courses based on soft skills. International Journal of AI in Education. DOI: 10.1007/s40593-025-00463-z

AI Content



	Text coverage	Words
AI Text	8.2%	399
Low Frequency		50
Medium Frequency		5
High Frequency		8
Human Text	91.8%	4
Excluded		
Omitted Words		0

About AI Detection

Our AI Detector is the only enterprise-level solution that can verify if the content was written by a human or generated by AI, including source code and text that has been plagiarized or modified. Learn more

AI Text

A body of text that has been generated or altered by AI technology. Learn more

Human Text

Any text that has been fully written by a human and has not been altered or generated by AI. Learn more

Copyleaks AI Detector Effectiveness

Credible data at scale, coupled with machine learning and widespread adoption, allows us to continually refine and improve our ability to understand complex text patterns, resulting in over 99% accuracy–far higher than any other AI detector–and improving daily. Learn more

Ideal Text Length

The higher the character count, the easier for our technology to determine irregular patterns, which results in a higher confidence rating for AI detection. Learn more

Reasons It Might Be AI When You Think It's Not

The AI Detector can detect a variety of AI-generated text, including tools that use AI technology to paraphrase content, auto-complete sentences, and more. Learn more

User AI Alert History

Historical data of how many times a user has been flagged for potentially having AI text within their content. Learn more

AI Logic

The number of times a phrase was found more frequently in AI vs human text is shown according to low, medium, and high frequency. Learn more

AI Logic

Shows you the "why" behind AI detection with sources you can see and verify.

AI Phrases

Detects phrases that appear with higher frequency in AI-written text than in human writing.

The frequency of a phrase in AI vs. human text.

3 x — 119x

119x of AI in Education,

How frequently the phrase was found in our dataset:

AI Text	**99.24 / 1,000,000 Documents**
Human Text	**0.84 / 1,000,000 Documents**

119x of AI in Education.

How frequently the phrase was found in our dataset:

AI Text	**99.24 / 1,000,000 Documents**
Human Text	**0.84 / 1,000,000 Documents**

64x New trends and research directions.

How frequently the phrase was found in our dataset:

AI Text	**1.06 / 1,000,000 Documents**
Human Text	**0.02 / 1,000,000 Documents**

36x Manyika, J., et al. (2017).

How frequently the phrase was found in our dataset:

AI Text	**3.14 / 1,000,000 Documents**
Human Text	**0.09 / 1,000,000 Documents**

30x Using NLP and ML.

How frequently the phrase was found in our dataset:

AI Text	**1.94 / 1,000,000 Documents**
Human Text	**0.06 / 1,000,000 Documents**

23x Jobs Lost, Jobs Gained: What the Future of Work

How frequently the phrase was found in our dataset:

AI Text	**1.39 / 1,000,000 Documents**
Human Text	**0.06 / 1,000,000 Documents**

14x natural language driven

How frequently the phrase was found in our dataset:

AI Text	**1.56 / 1,000,000 Documents**
Human Text	**0.11 / 1,000,000 Documents**

14x McAfee, A. (2014). The Second Machine Age.

How frequently the phrase was found in our dataset:

AI Text	**1.71 / 1,000,000 Documents**
Human Text	**0.13 / 1,000,000 Documents**

12x Will Mean for Jobs, Skills, and Wages.

How frequently the phrase was found in our dataset:

AI Text	**1.37 / 1,000,000 Documents**
Human Text	**0.11 / 1,000,000 Documents**

10x Skill gap analysis

How frequently the phrase was found in our dataset:

AI Text	**6.69 / 1,000,000 Documents**
Human Text	**0.67 / 1,000,000 Documents**

9x JSON Web Token (JWT).

How frequently the phrase was found in our dataset:

AI Text	**17.74 / 1,000,000 Documents**
Human Text	**2.07 / 1,000,000 Documents**

6x overview of cloud computing

How frequently the phrase was found in our dataset:

AI Text	**5.37 / 1,000,000 Documents**
Human Text	**0.87 / 1,000,000 Documents**

3x data mining to predict

How frequently the phrase was found in our dataset:

AI Text	**1.01 / 1,000,000 Documents**
Human Text	**0.33 / 1,000,000 Documents**

SKILLGAP AI: A SMART SKILLING ASSESSMENT AND INDIVIDUALIZED COURSE RECOMMENDATION SYSTEM

Tanisha
Chitkara University Institute of Engineering and Technology
Chitkara University
Punjab, India
tanisha2440.be22@chitkara.edu.in

Tanishq Choudhary
Chitkara University Institute of Engineering and Technology
Chitkara University
Punjab, India
tanishq2445.be22@chitkara.edu.in

Tanmay Gupta
Chitkara University Institute of Engineering and Technology
Chitkara University
Punjab, India
tanmay2449.be22@chitkara.edu.in

Susen Narayan
Chitkara University Institute of Engineering and Technology
Chitkara University
Punjab, India
susen2428.be22@chitkara.edu.in

Swati Goel
Assistant Professor, Chitkara University Institute of Engineering and Technology
Chitkara University
Punjab, India
swati.goel@chitkara.edu.in

***Abstract*—The absence of knowledge about what skills are lacking and which courses to take to improve them is one of the most frequent problems learners face today. The majority of online platforms bombard users with a list of courses without identifying what that particular individual actually needs. This paper presents SkillGap AI — a platform designed to address the problem through skill-based analysis and personalized course recommendation. The system first takes user-provided skill inputs and applies machine learning techniques to provide the career paths. Depending on their input, it identifies which areas they are weak in and then suggests the corresponding courses from various existing platforms. A career dataset of 4,077 records is used to train and evaluate seven machine learning classifiers with TF-IDF vectorization. The SGD Classifier achieves the highest accuracy of 94.48**

***Index Terms*—Skill Gap Analysis, Course Recommendation, Machine Learning, TF-IDF, Skill India Digital Hub, SGD Classifier, Random Forest, Node.js, MongoDB, Personalized Learning, Workforce Development, NLP**

I. Introduction

The shift of India towards digital growth has created a massive issue in which the skill set which individuals hold and the skill set which employers require has started to diverge even further. According to the National Skill Development Corporation, over 400 million workers in India would need re-skilling by the year 2030 due to the rapid changes being

brought about by technology in the job market [1]. Not withstanding these facts, even with access to more than a dozen online learning platforms, most people find it very difficult to determine what exactly they need to learn. They browse through hundreds of learning possibilities and often fail to complete a course because it was not exactly what they needed. One such platform is the Skill India Digital Hub, run by the Ministry of Skill Development and Entrepreneurship. It is one of the most helpful resources available, offering over 752 free and industry-accredited courses developed in association with partners including IBM, Microsoft, Cisco, HCL, IGNOU, and UpGrad [2]. The limitation is that even SIDH cannot decide which specific course a learner should take. This is precisely what SkillGap AI aims to solve. With the help of SkillGap AI, students get to do an accurate assessment within their domain to know the areas they are lacking in and then directly directed to SIDH course(s) that cover them. All these assessments will be persistent; therefore, whenever they wish, they can come back and redo the assessment to see whether they have really made improvements or not [3]. Personalization and adaptive feedback have been identified as some of the biggest unsolved problems of online education through the research carried out on educational data mining [3]. Segmentation of users based on their skills to recommend content to them is known to make the recommendations more relevant [4]. High accuracy, reaching up to 99.2 • Lack of personalized skill gaps: Most portals make course recommendations based on general categories and trending courses, rather than focusing on specific gaps that the learner may have. • Lack of correlation to government's free courses: The learners lack awareness of

the 750+ free and certified courses available at SIDH that can help fill the gaps identified by the learners. • Lack of measurable improvements: Existing portals lack the ability to enable learners to re-take the test and measure their overall skill improvements. The main contributions and features of this research paper are: • A system that integrates the assessment process with India's national government course catalog (SIDH) — something not available on any existing platform. Learners' ability can be tracked and their progress can be monitored through repeated test attempts. • A full-fledged backend implemented in Node.js, Express, and MongoDB to handle authentication, test administration, domain-wise scoring, skill gap identification, and result storage. • Evaluation of seven different machine learning classifiers using features derived via the TF-IDF algorithm on a real career dataset, where the SGD Classifier achieved the highest accuracy of 94.48

• All government-approved courses are linked on a single platform, providing free and accessible recommendations for all learners across India. The paper is organized as follows: Section II reviews the related work across four domains relevant to SkillGap AI. Section III presents the combined system architecture and methodology, covering the assessment design, gap detection, TF-IDF career prediction, and course recommendation. Section IV describes the datasets used in this study. Section V presents the results and discussion including model performance comparison, longitudinal improvement, and comparative analysis. Section VI discusses the challenges and limitations. Section VII concludes the paper with future scope.

II. Related Work

This section reviews literature from five different domains that have a direct connection with SkillGap AI: educational data mining, skill gap identification, course recommendation systems, text classification, and career prediction through machine learning.

A. Educational Data Mining & Learning Analytics

A study by Aldowah et al. [3] presented a detailed review of EDM and learning analytics research and highlighted that personalization and adaptive feedback have remained two persistently unsolved problems. Many systems monitor learner performance, but very few leverage that data to nudge the learner towards the correct path. The work by Kausar et al. [4] further established that segmenting learners based on skill levels before suggesting content greatly enhances recommendation relevance — the core concept driving our ML module. A related study by Uzoma et al. [5] presented a machine learning recommender system for student course selection using Random Forest and Decision Trees, obtaining accuracies as high as 99.2

B. Course Recommending Systems

Classification into three categories of recommenders was done by Burke [6]: content-based filtering, collaborative filtering, and hybrid approach, together with a detailed discussion of their applicability in the education field. An automated approach for a recommendation system based on hybrid filtering was proposed by Sharma et al. [7]; it provided more reliable outcomes than other methods. In their systematic review conducted in the period from 2015 till 2022 [8], Silva et al. noted the absence of educational recommender systems incorporating real-time evaluation of skills with a suggested course – precisely what SkillGap AI is supposed to do. Another research conducted at ACM SIGIR 2024 [9] revealed that labor market data also have to be considered.

C. Skill Gap Analysis & Career Prediction

It is evident that skill gaps play a significant role economically due to the work done by Manyika et al. [10] and Brynjolfsson & McAfee [11]. In the Indian scenario, Roy & Singh [12] identified skill gaps through sectors using NSDC data without designing an automated process to address such gaps. An analysis was performed recently on career predictions based on natural language processing and involved training models using KNN, SGD, Logistic Regression, Random Forest, and Adaptive Boosting algorithms using data on 2,000 students and identified KNN and SGD as having high accuracy, in line with our results.

D. TF-IDF and Text Classification for Career Matching

According to Das et al., the use of both TF-IDF weight- ing and Random Forest algorithms proved most effective for text classification, with an accuracy rate of 93.81%. In all the literatures reviewed, there has been no existing approach in which all four factors, which include structure domain assessment, machine learning gap analysis, TF-IDF skill text classification, and instant course recommendation based on a DPI national directory curated by the government, are incorporated.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

The Skill Gap AI application system is designed as a modular web-based application. Every module executes one unique function and works independently of other modules. This part of the paper presents the architecture of the Skill Gap AI application and describes the entire methodological process of using the application, starting from logging in all the way to course recommendation. This process is represented graphically in the flowchart presented in Fig. 1.

A. User Interface and Assessment Layer

The frontend, whose design is ongoing, will provide a user-friendly interface with registration, login, test-taking, and result-viewing functionality. The API endpoints for the test have been created and are functioning correctly. Questions in the question bank are grouped into seven skill domains: Programming Fundamentals, Data Literacy, Cloud Computing, Cybersecurity, Communication Skills, Digital Productivity,

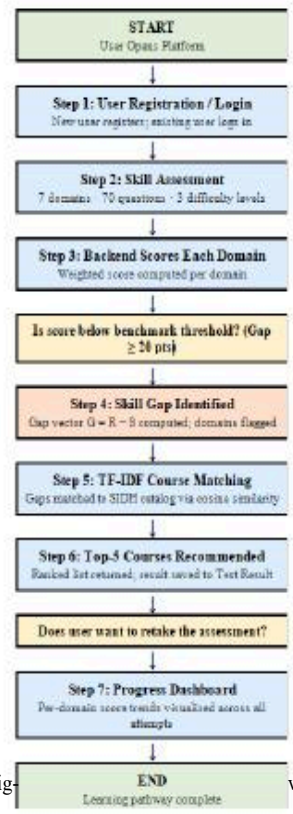


Fig- wchart

and Domain-Specific Technical Skills. Questions are tagged by skill area and labeled as easy, medium, or hard to enable weighted scoring.

B. Scoring and Gap Identification

- When a test is submitted, the backend assesses the learner's performance separately for each domain. Scores are assigned for each of the seven domains and then compared against benchmark scores derived from actual developer survey data [16]. If a learner's score in any domain is substantially below the benchmark, a skill gap is identified. Each student is represented as a feature vector as shown in equation (1):

$$S = [Age, Education, Skills, Interests](1) \quad (1)$$

Label Encoding converts each categorical field to an integer. For Education: Bachelor's = 0, Master's = 1, PhD = 2. The gap for each dimension is defined by equation (2):

$$G_i = R_i - S_i, \quad \text{where } i \in \{\text{Age, Education, Skills, Interests}\} \quad (2)$$

For example, for a student $S = [25, 1, 2, 1]$ with the Software Engineer benchmark $R = [25, 2, 3, 1]$, the gap vector $G = [0, 1, 1, 0]$ — meaning the learner has gaps in Education and Skills domains. The gap vector $G = [G_1, G_2, \ldots, G_7]$ is computed across all seven domains. Only domains where $G_i > 0$ are flagged. Domains where $G_i \leq 0$ indicate the learner meets or exceeds the benchmark and require no intervention.

C. TF-IDF Based Career Prediction

A career prediction model is built using TF-IDF vectorization on the Career Dataset. Skills data is first converted to lowercase, then passed to the TF-IDF vectorizer with ngram_range=(1,3), max_df=0.85, and stop_words='english'. The TF-IDF weight for each feature term is given by equation (3):

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log \frac{N}{DF(t)} \quad (3)$$

Here, TF(t,d) is the frequency of term t in document d, N is the total number of records, and DF(t) is the number of records containing term t. A sparse feature matrix is produced comprising both individual skill terms and multi-word phrases such as 'machine learning' and 'deep learning nlp'. This feature matrix is used to train seven classifiers: Decision Tree, Logistic Regression, Naive Bayes, Random Forest, SGD Classifier, KNN, and Ridge Classifier.

D. Recommendation Module

The recommendation module takes the list of gap domain flags and matches them against the SIDH course catalog. Each course is represented as a TF-IDF vector of its domain tags and description keywords. Cosine similarity between the learner's gap-domain query vector Q and each course vector c is computed using equation (4):

$$\text{similarity}(Q, c) = \frac{Q \cdot c}{\|Q\| \, \|c\|} \quad (4)$$

The module returns the top 5 courses with the highest similarity scores. This is a transparent model: users can clearly understand which skill terms contributed to the recommendations, and no historical user data is required to operate it.

E. Storage and Progress Management

Every test attempt is recorded in the TestResult collection: domain scores, gap values, recommended courses, and submission timestamp. This enables the progress dashboard to plot per-domain improvement trends across multiple test cycles, implementing the "assess, learn, reassess" feedback loop that characterizes self-regulated learning theory [17].

F. Performance Evaluation Metrics

All classifiers are evaluated using four standard metrics computed from the confusion matrix:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

$$\text{F1-Score} = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \quad (8)$$

where TP denotes True Positive, TN denotes True Negative, FP denotes False Positive, and FN denotes False Negative. Metric values are computed per class and averaged across all five career classes (macro average).

G. Implementation Details

Implementation and Experimental Setup

The suggested research aims at implementing a system based on machine learning algorithms, that would predict the type of profession and suggest courses according to certain skills. For this purpose, the experiments were conducted using Python and associated libraries.

- Dataset Creation
  The dataset used in this research includes 4,077 entries and comprises two features, which include Career and Skill. As each entry correlates the certain set of skills with a career, the preparation procedure involves lowering the case, deleting duplicates, and normalizing the labels of career types.
- Features Extraction
  As the data is comprised of texts, it was decided to extract the features using the TF-IDF vectorization. It includes such parameters as n-grams range of (1,3), removing common stop words, and applying maximum document frequency to filter the text features.
- Modeling and Training
  For training purposes, the prepared dataset was divided using a stratified split method in the ratio of 80:20. There were multiple machine learning models applied for the task, including decision tree, logistic regression, naive bayes, random forest, SGD classifier, k-nearest neighbors (knn), and ridge classifier.
- Career Prediction
  For each model developed, the predicted career path is suggested based on the user's skills inputted into the model. Among all the models tested, the SGD Classifier has shown the highest efficiency in dealing with high-dimensional sparse TF-IDF output.
- Skills Gaps Identification
  In order to detect skill gaps, a rules-based technique is used to compare skills specified by the user to pre- defined sets of skills required for each profession. The discrepancy is determined as a gap.
- Course Suggestion
  Based on the detected gaps, specific courses are selected for recommendation according to a predefined skill-to-resource map.
- Evaluation Metrics
  Performance of the models is analyzed using the following measures : Accuracy, Precision, Recall, F1 score. Confusion matrix was also generated to analyze model performance for each class.

Dataset	Source	How We Use It
Stack Overflow Developer Survey	Stack Overflow	Establish domain-specific threshold standards for proficiency
Online Course Recommendation Dataset	Kaggle / UCI	Training and testing of the course recommendation engine
Student Performance Dataset	UCI ML Repository	Test scoring methodology and effectiveness of detecting gaps
Workforce / Digital Skills Survey	ITU / NSDC	Career recommendations according to job market demand
AI-Based Career Recommendation Dataset	Kaggle	Test feature encoding for career prediction (200 entries, 32 careers)
Career Dataset	Excel / Custom	Training and testing of 7 ML classifiers (2,084 entries)

TABLE I
SKILLGAP AI DATASETS

H. DATASETS AND DATA SOURCES

Six datasets are used to design, calibrate, train, and evaluate the system. Table 1 summarises all six datasets and their respective roles in the system.

IV. DATASETS AND DATA SOURCES

Domain benchmark scores are derived from the Stack Overflow Developer Survey [16], which collects responses from over 90,000 software engineers globally each year. Rather than setting arbitrary thresholds, we use median proficiency levels of actual working professionals to calibrate gap detection.

A. Stack Overflow Developer Survey

Domain benchmark scores are derived from the Stack Overflow Developer Survey [16], which collects responses from over 90,000 software engineers globally each year. Rather than setting arbitrary thresholds, we use median proficiency levels of actual working professionals to calibrate gap detection.

B. Online Course Recommendation Dataset

A Kaggle dataset of learner-course interaction logs is used to pre-train the recommendation module. It contains historical data about which courses learners enrolled in, how far they progressed, and how they rated them.

C. Student Performance Dataset

The UCI Student Performance Dataset [19] provides academic performance records across multiple subjects and demographic attributes. It is used to validate that our weighted scoring scheme captures genuine competency differences rather than measurement noise.

D. Workforce and Digital Skills Survey

Workforce demand data from ITU and NSDC [1] is applied to supplement course ranking by prioritising recommendations in skill domains currently in high demand among employers.

E. AI-Based Career Recommendation Dataset

This dataset includes 200 student records with fea- tures {Age, Education, Skills, Interest} and target Recom- mended_Career. Recommendation_Score ranges from 0.85 to 0.95 (mean = 0.896). The 80/20 split gives 160 training and 40 test records, used to validate the feature encoding and gap detection logic.

F. Career Dataset

The Career Dataset (Excel, 4,076 entries, 2 columns: Career and Skill) covers six career categories post label consolidation: Artificial Intelligence, Data Science, Development, Security, Software Development, and Engineering. After duplication, 2,084 clean records remain. The stratified 80/20 split gives 1,667 training and 417 test entries. This is the primary dataset used to train and evaluate all seven classifiers.

RESULTS AND DISCUSSION

G. Classifier Evaluation

Seven classifiers were trained and evaluated on the Career Dataset (2,084 records, six career classes) using TF-IDF features with trigram support and an 80/20 stratified split. Before presenting the results, we briefly describe each classifier and how it works in the context of this task.

- Decision Tree: A tree structure method which predicts career choices based on decision rules applied sequentially. The nodes within this model are skill feature conditions whereas the leaves are predicted careers. This technique is easily understandable. Accuracy: 88.73
- Logistic Regression: Probability based linear classifier that determines the probability of each class of occupation using soft max and then assigns to the class that has the maximum probability. This model performs very well on high-dimensional TF-IDF vectors. Accuracy: 93.29
- Naive Bayes: It is a probabilistic classifier derived from the Bayes theorem where it assumes that all attributes are independently distributed. This model will classify the job type with the maximum posterior probability based on the observed terms in the skills. Fast performance in textual classification and its accuracy: 83.21
- Random Forest: A combination of 200 decision trees that individually vote for the prediction of career; the result comes from the majority vote by the decision trees. It improves upon the accuracy of an individual Decision Tree by reducing its over fitting and variance. Accuracy: 93.05
- SGD Classifier (Best Model): The Stochastic Gradient Descent Classifier is a linear classifier which learns by incrementally updating its weights using the difference between the prediction error and the actual output after processing each individual data point. The algorithm is very effective in sparse TF-IDF matrices and results in the convergence to a powerful linear decision boundary. It attained the highest accuracy of 94.48
- KNN: Classifier with cosine/Euclidean nearest neighbor search for the 5 closest neighbors in the training set and prediction based on the career category present in the majority of the 5 closest training samples. Training-free, but expensive due to high dimensionality of the TF-IDF feature vectors. Accuracy: 89.69
- Ridge Classifier: L2 Regularized Linear Classifier that penalizes large weights and helps prevent overfitting, giving better generalization capabilities on new/unseen samples. The accuracy is very close to the SGD classifier, with 93.76

TABLE II
CLASSIFIER PERFORMANCE — ALL SEVEN CLASSIFIERS ON CAREER DATASET

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	88.73%	78.65%	83.33%	80.92%
Logistic Regression	93.29%	85.71%	92.86%	89.14%
Naive Bayes	83.21%	89.04%	77.38%	82.80%
Random Forest	93.05%	84.62%	91.67%	88.00%
SGD Classifier	94.48%	88.64%	92.86%	90.70%
KNN	89.69%	78.49%	86.90%	82.49%
Ridge Classifier	93.76%	89.53%	91.67%	90.59%

The SGD Classifier provides the best accuracy score (94.48

Figures 2–6 illustrate the performance plots based on the evaluation score

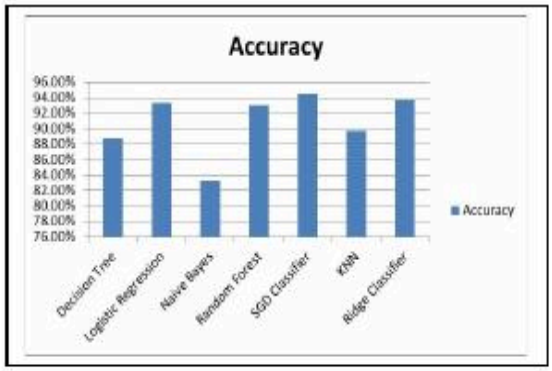


Fig. 2. Accuracy Comparison

In the graph above, accuracy is plotted for all classifiers. The SGD classifier achieves the highest accuracy of 94.48%, followed by Ridge (93.76%) and Logistic Regression (93.29%). Naive Bayes records the lowest accuracy at 83.21%.

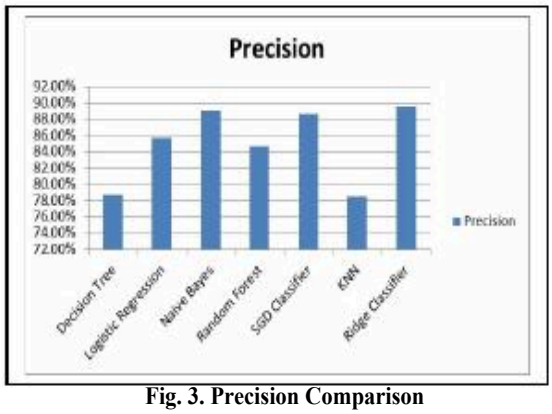


Fig. 3. Precision Comparison

Naive Bayes has a high precision rate at 89.04% due to conservative prediction. The other two classifiers with higher precision rates are Ridge and SGD above 88%. The two classifiers having the lowest precision rates below 78% are Decision Tree and KNN.

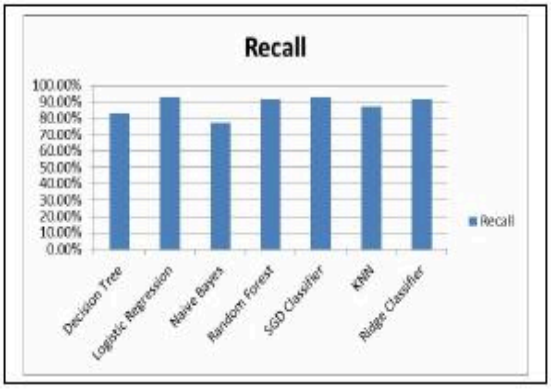


Fig. 4. Recall Comparison

Naive Bayes has low recall at 77.38%. This is evidence that it does not catch many positive predictions. The classifier with the best recall levels above 92.8

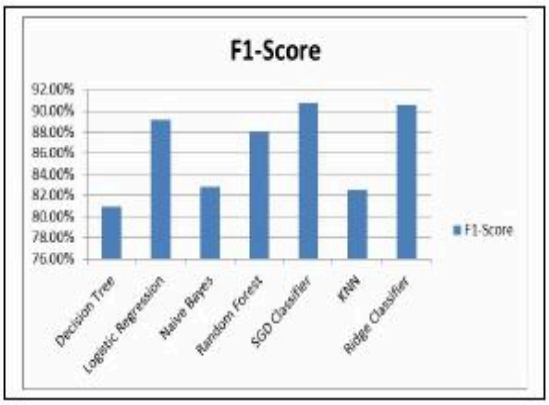


Fig. 5. F1-Score Comparison

SGD provides the highest F1-Score (90.70%) value, followed by Ridge (90.59%). Decision Tree and KNN have the lowest F1-Scores, which suggests poor precision and recall rates.

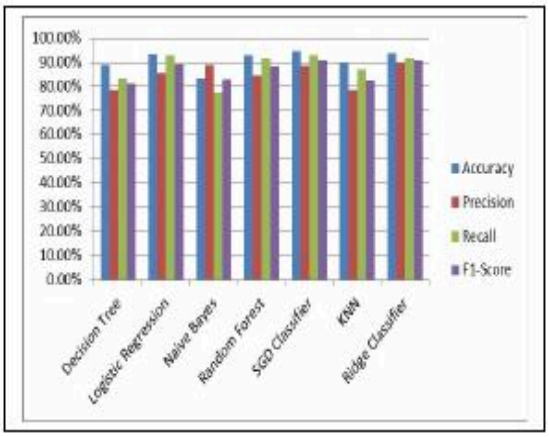


Fig. 6. Combined Performance Comparison

Grouped bar charts include all four measures. Linear models perform better than other models, whereas Naive Bayes exhibits clear asymmetry between high precision and low recall rates.

Figures 7 to Figure 13 illustrate confusion matrices for seven algorithms. Rows indicate the true labels, whereas columns indicate predicted values. Labels: 0=AI, 1=Data Science, 2=Security, 3=Software Development, 4=UX/UI Design.

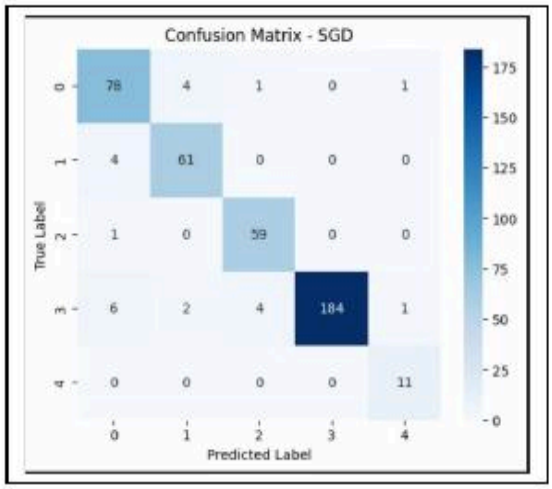


Fig. 7. Confusion Matrix — SGD Classifier (Best, Accuracy: 94.48%)

Diagonal dominance is very strong. Class 3 (Software Dev., 184/197 accurate) and Class 4 (UX/UI, 11/11) perform best. Errors include 6 cases in Class 3 wrongly predicted as Class 0.

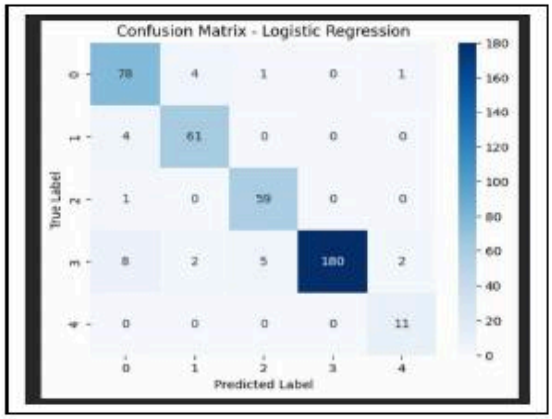


Fig. 8. Confusion Matrix — Logistic Regression (Accuracy: 93.29%)

Comparable performance to stochastic gradient descent. Error involves 8 cases in Class 3 wrongly predicted as Class 0. Class 4 is completely accurately predicted (11/11).

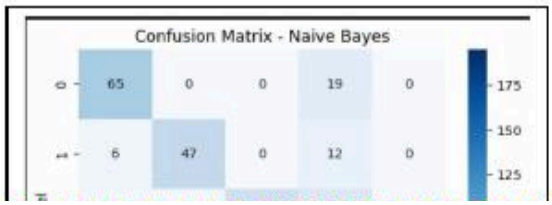


Fig. 9. Confusion Matrix — Naive Bayes (Accuracy: 83.21%)

Pattern exists where high percentages of Classes 0, 1, and 2 are wrongly predicted as Class 3 (Software Dev.).

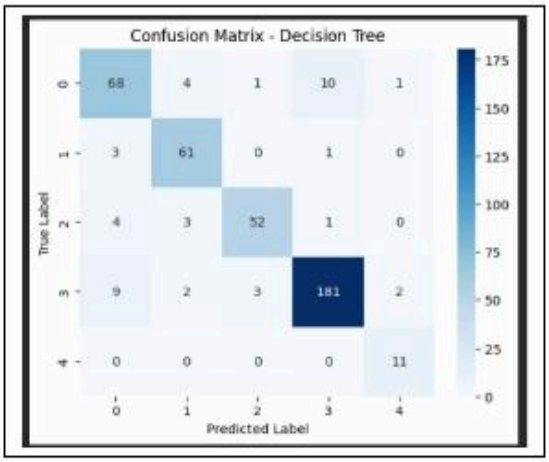


Fig. 10. Confusion Matrix — Decision Tree (Accuracy: 88.73%)

Significant error: 10 Class 0 instances classified as Class 3, and 9 Class 3 instances classified as Class 0.

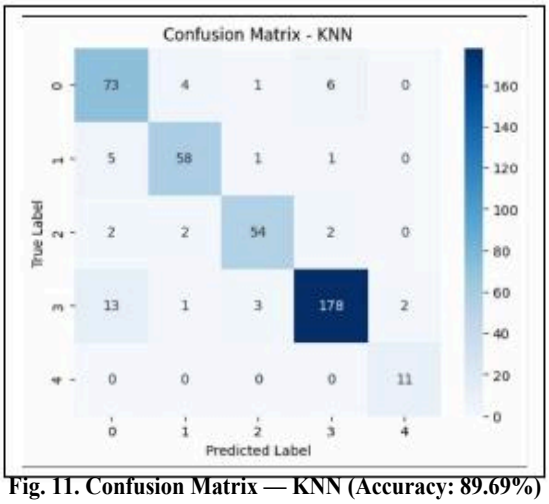


Fig. 11. Confusion Matrix — KNN (Accuracy: 89.69%)

Major error: 13 Class 3 instances incorrectly classified as Class 0. Class 4 is accurately classified (11/11).

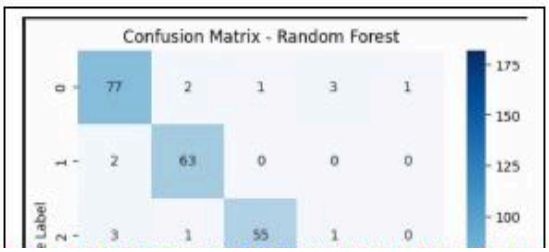


Fig. 12. Confusion Matrix — Random Forest (Accuracy: 93.05%)

Excellent classification performance. Errors are mostly in Class 3 (9 instances classified as Class 0). Ensemble mitigates overfitting compared to one Decision Tree.

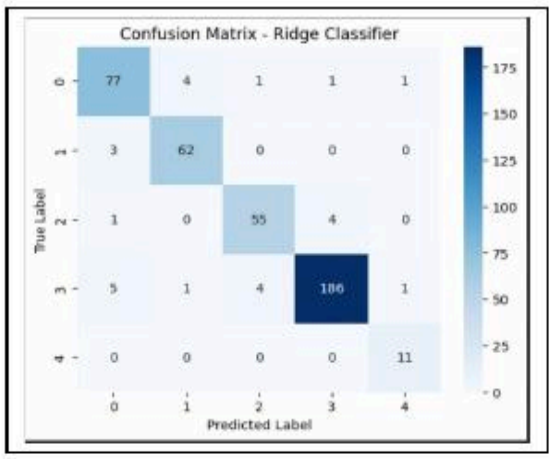


Fig. 13. Confusion Matrix — Ridge Classifier (Accuracy: 93.76%)

Similar to SGD.L2 regularization avoids overfitting. Errors include 5 Class 3 instances classified as Class 0 and 4 Class 2 instances classified as Class 3.

H. Classifier Performance

Seven machine learning classifiers were trained and evaluated on the Career Dataset consisting of 2,084 records across five career classes. TF-IDF vectorization with trigram features was applied, and an 80/20 stratified train-test split was used. The evaluated classifiers include Decision Tree, Logistic Regression, Naive Bayes, Random Forest, SGD Classifier, KNN, and Ridge Classifier.

Among all models, the SGD Classifier achieved the highest performance with an accuracy of 94.48

V. CHALLENGES AND LIMITATIONS

The implementation of this system has raised several issues. Firstly, there is currently no publicly available API for SIDH, thus the scraping of course information at regular intervals from the public listing is necessary. While this solution works for the time being, it cannot be relied upon in the long term; the integration of an official API will provide real-time catalog updates. The

evaluation conducted in this paper involves seven skill categories, which is sufficient to begin with; however, the real-world scenario encompasses far more than that. Inclusion of additional questions covering soft skills, certifications relevant to specific domains, and emerging tech topics should be considered in future versions of the system. A cold-start issue also exists, wherein new users are given a single test instance on which recommendations can be based, failing to accurately capture their true set of skills. The problem of class imbalance in the Career Dataset is prevalent, especially for minority classes like UX/UI Design (54 instances vs. 987 for Software Development). Future work will involve exploring ways of handling imbalance using approaches like oversampling through SMOTE and balancing of class weights. Finally, ensuring fairness among all questions regardless of educational background, languages, and regionsin India requiresa careful manual audit before wide-scale deployment.

VI. CONCLUSION AND FUTURE SCOPE

The Skill Gap AI was built with the premise that most people do not know what skills they are missing, and most platforms do not provide them with this information. Learners are able to ascertain their status first before being guided towards relevant Skill India Digital Hub courses. The back end is fully built and functional, covering all aspects such as user management, test workflow, domain-based scoring, skill gap determination, and course management. Based on empirical results from the application of seven machine learning classifiers on the Career dataset, the SGD Classifier performs best and predicts skill gaps with an accuracy rate of 94.48Future research will look into generating questions based on the learner's needs through LLMs, employing reinforcement learning for dynamic course ranking, developing a formal API interface between SkillGap AI and Skill India Digital Hub, and adding more domains to the Skill India Digital Hub catalog.

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REFERENCES

[1] NSDC (2023). India Skills Report 2023. Ministry of Skill Development and Entrepreneurship, Government of India.
[2] SIDH. Skill India Digital Hub. Course Catalog and Platform Overview. Ministry of Skill Development and Entrepreneurship. https://www.skillindiadigital.gov.in
[3] Aldowah, H., Al-Samarraie, H., & Fauzy, W. M. (2019). Educational Data Mining and Learning Analytics for 21st Century Higher Education. Telematics and Informatics, 37, 13–49.
[4] Kausar, S., Huahu, X., Hussain, I., Wenhao, Z., & Zahid, M. (2018). Integration of Data Mining Clustering Approach in the Personalized e- Learning System. IEEE Access, 6, 72724–72734.
[5] Uzoma, T. C., Ikerionwu, C. O., Nwanga, M. E., & Etus, C. (2024). Machine learning recommender system for an enhanced students course selection. International Journal of Computer Trends and Technology, 72(10), 31–35.
[6] Burke, R. (2002). Hybrid Recommender Systems: Survey and Experiments. User Modeling and User-Adapted Interaction, 12(4), 331–370.
[7] Sharma, S., Rana, V., & Malhotra, M. (2022). Automatic recommendation system based on hybrid filtering algorithm. Education and Information Technologies, 27(2), 1523–1538
[8] Silva, F. L., et al. (2023). A systematic literature review on educational recommender systems for teaching and learning. Education and Information Technologies, 28(3), 3289–3328.
[9] Bothmer, K., & Schlippe, T. (2024). Course recommender systems need to consider the job market. Proceedings of the 47th ACM SIGIR Conference, pp. 1–10.
[10]] Manyika, J., et al. (2017). Jobs Lost, Jobs Gained: What the Future of Work Will Mean for Jobs, Skills, and Wages. McKinsey Global Institute.
[11] Brynjolfsson, E., & McAfee, A. (2014). The Second Machine Age. W.W. Norton & Company.
[12] Roy, A., & Singh, P. (2022). Skill gap analysis in the Indian IT sector using NSDC survey data. Journal of Vocational Education and Training, 74(3), 412–430.
[13] Casuat, C. D., et al. (2024). Unlocking futures: A natural language driven career prediction system. arXiv preprint arXiv:2405.18139.
[14] Das, M., Kamalanathan, S., & Alphonse, P. J. A. (2023). A comparative study on TF-IDF feature weighting method and its analysis using unstructured dataset. arXiv preprint arXiv:2308.04037.
[15] Abdulkader, A., et al. (2024). Automated Classification and Recommendation of Research Articles Using NLP and ML. arXiv preprint arXiv:2510.05495.
[16] Stack Overflow. (2023). Stack Overflow Annual Developer Survey 2023. https://survey.stackoverflow.co/2023
[17] Zimmerman, B. J. (2002). Becoming a self-regulated learner: An overview. Theory into Practice, 41(2), 64–70.
[18] Jones, M., Bradley, J., & Sakimura, N. (2015). JSON Web Token (JWT). RFC 7519. IETF.
[19] Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. Proceedings of FUBUTEC, Porto, Portugal, pp. 5–12.
[20] Drachsler, H., & Greller, W. (2016). Privacy and Learning Analytics — It's a DELICATE Issue. LAK 2016 Proceedings, Edinburgh, pp. 89–98.
[21] Zhang, S., Yao, L., Sun, A., & Tay, Y. (2019). Deep Learning-based Rec- ommender System: A Survey and New Perspectives. ACM Computing Surveys, 52(1), 1–38
[22] Ali, A. (2022). An overview of cloud computing for the advancement of the e-learning process. Journal of Theoretical and Applied Information Technology, 100(3), 847–855.

[23] Ricci, F., Rokach, L., & Shapira, B. (2015). Recommender Systems Handbook (2nd ed.). Springer.
[24] Marras, M., Boratto, L., Ramos, G., & Ghidini, C. (2022). Equality of learning opportunity via individual fairness in personalized recommendations. International Journal of AI in Education, 32, 636–684.
[25] Dunn, J. (2021). Adaptive testing: A review and guide for practitioners. Educational Measurement: Issues and Practice, 40(2), 5–22.
[26] Varghese, B., & Buyya, R. (2018). Next generation cloud computing: New trends and research directions. Future Generation Computer Systems, 79, 849–861.
[27] Ministry of Skill Development and Entrepreneurship (MSDE). (2024). Annual Report 2023–24. Government of India.
[28] Marras, M., et al. (2025). A recommender system of postgraduate courses based on soft skills. International Journal of AI in Education. DOI: 10.1007/s40593-025-00463-z